# Casey Doe

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Hillcrest, ST | Willing to relocate

## Experience

**Hardware Engineering Intern, Example Avionics**, Northgate, ST — May 2025 – Aug 2025

- Cut noise on a 16-bit ADC front end from 12 LSB to 3 LSB RMS by redesigning the anti-aliasing filter as a 2nd-order active stage and rerouting its ground return in Altium, checked with an FFT of 1 million samples
- Found 7 layout faults during bring-up of 3 sensor board revisions with an oscilloscope and a thermal camera
- Wrote a Python harness to check 64 signals over SPI in 90 s, which production used on the first 200 units
- Sized an LC filter in LTspice for the 3.3 V rail and saw 60% less switching noise on a spectrum analyzer

**Embedded Software Intern, Example Robotics**, Hillcrest, ST — May 2024 – Aug 2024

- Ported STM32 motor-control firmware from bare-metal C to FreeRTOS and kept 10 kHz control loop jitter under 2 µs by running the loop from a timer interrupt outside the scheduler, measured on a logic analyzer
- Added a CAN bootloader that flashes new firmware in 40 s, with no failures in 100 updates
- Fixed an encoder counter overflow that put 1 in 500 moves off position, then ran 24 hours with no errors

**Undergraduate Research Assistant, Example State Power Lab** — Sep 2023 – Apr 2024

- Measured switching losses on 12 GaN transistors with a double-pulse tester, all within 5% of the datasheet
- Shortened test sessions from 2 hours to 30 minutes with an FPGA pulse generator in Verilog and MATLAB

**Tutor, Example State Learning Center** — Jan 2023 – May 2023

## Projects

**Senior Design: Battery Management System for an Electric Kart (Altium, STM32, C, CAN, LTspice)** — Fall 2025 – Spring 2026

- Kept a 24-cell Li-ion pack balanced within 10 mV over 30 charge cycles with an STM32 that polls 2 cell-monitor ICs over SPI every 100 ms and switches 24 bleed resistors on a custom 4-layer Altium board
- Wrote a 10 Hz thermal model in C that estimates cell temperature within 1.5 °C of thermocouple readings
- Shut the pack down within 5 ms of an overcurrent fault in 25 out of 25 short-circuit tests

**Software-Defined Radio FM Receiver (Python, GNU Radio, RTL-SDR, NumPy)** — Spring 2025

- Demodulated stereo FM in real time at 2.4 MS/s in GNU Radio on Linux and tested it on 6 local stations
- Cut CPU use 40% by decimating earlier and replacing 2 blocks with a NumPy polyphase filter
- Measured 55 dB stereo separation with a test tone, 5 dB short of a commercial receiver on the same antenna

**Self-Balancing Robot (Arduino, C++, MPU-6050, MATLAB, KiCad)** — Fall 2024

- Tuned a PID loop in MATLAB that kept a 2-wheel robot upright for over 10 minutes, even after 15° pushes
- Got tilt error under 1° against an inclinometer with a complementary filter on MPU-6050 data
- Replaced the breadboard with a 2-layer motor driver board in KiCad, which had no wiring faults in 40 runs

## Education

**B.S. Electrical Engineering, Minor in Computer Science**, GPA 3.88/4.00 — Spring 2026
Example State University
**Relevant coursework:** Embedded Systems, Digital Signal Processing, Introduction to High Power Electronics, Modern Control Design, Passive RF and Microwave Devices, Computer Organization, Introduction to VLSI Design
**Activities:** IEEE student branch, treasurer, Tau Beta Pi, Formula SAE electrical team

## Skills

**Software:** C, C++, Python, MATLAB, Verilog, Altium, KiCad, LTspice, FreeRTOS, GNU Radio, Linux
**Hardware/Lab:** Oscilloscope, Spectrum analyzer, Logic analyzer, Double-pulse tester, Thermal camera, STM32, FPGA
**Certifications:** Engineer in Training (EIT), IPC J-STD-001 Soldering